RADAR RESEARCH PORTFOLIO

Object Tracking, DSP Firmware, SiL Replay, and Distributed Simulation

TECHNICAL CONCEPT COMPENDIUM | BASELINE VS PROPOSED MATHEMATICS | MECHANISMS | NUMERICAL EXAMPLES

24 research concepts across 4 repositories

Repository 1: Central Object Tracker
`CORECOMP/ALSW/OT_ObjectTracking`

Repository 2: Satellite Radar DSP Firmware
`Core_Radar_Gen8_iND13400`

Repository 3: Domain Controller SiL Replay Harness
`Core_RESIM_DC_Emb_Library`

Repository 4: Distributed Simulation and Cluster Infrastructure
`Core_RESIM_HPCC / Engine`

Prepared from the supplied research notes. Performance figures are reported values and should be validated against repository tests, logs, and KPI definitions before release.

Executive Summary

This document reorganizes the supplied material into an engineering research portfolio. Each concept is expressed as a before/after formulation, an operational mechanism, a physical or numerical example, implementation checkpoints, and a principal technical risk.

Portfolio thesis: Use physical invariants, uncertainty-aware estimation, robust statistics, and parallel algebra to improve radar accuracy while reducing latency and compute cost.

Repository	Concepts	Primary focus	Representative outcome
Central Object Tracker	8	Object tracking, multipath physics, association robustness, and uncertainty-aware filtering	Tracking RMSE 1.969 m to 0.651 m
Satellite Radar DSP Firmware	7	Embedded radar calibration, CFAR, prioritization, and low-compute estimators	Alignment lock <= 2.3 s
Domain Controller SiL Replay Harness	4	Deterministic replay, asynchronous sensor deskew, and calibration invariants	Fusion error 0.68 m to 0.09 m
Distributed Simulation and Cluster Infrastructure	5	Radar-native simulation, parallel state estimation, synthetic augmentation, and replay certification	50,000-step scan in 16 levels

Cross-cutting design patterns

- Virtual sensing and specular invariants turn multipath from interference into geometry.
- Adaptive and distribution-free gates preserve tracks during maneuvers without uncontrolled clutter admission.
- Track-conditioned estimators trade expensive global transforms for small residual calculations.
- Associative operators and state lifting remove serial dependencies from large-scale replay.
- Replay correctness is treated as a certifiable property rather than an empirical warm-up choice.

Evidence and validation note

The numerical outcomes in this report are transcribed from the supplied content. They are not independently verified here. For an engineering release, every metric should be linked to a dataset, commit, configuration, KPI script, confidence interval, and reproducible command.

Portfolio Map

Repository 1: Central Object Tracker

`CORECOMP/ALSW/OT_ObjectTracking`

01. Virtual Aperture Specular Tracking `[Msmt_Update_ObjTrks_CCA_Moveable.cpp]`
02. Rank-1 Specular Invariant and Bearing Correction `[f360_detect_multipath.cpp]`
03. Kinematics-Aware Adaptive Gating `[f360_determine_association_hypothesis.cpp]`
04. Tangent-Mirror Hermite Linearization `[f360_is_reflective_guardrail_track.cpp]`
05. Spectral Entropy Covariance Scaling `[Msmt_Update_ObjTrks_CTCA.cpp]`
06. Distribution-Free Conformal Gating `[f360_determine_association_hypothesis.cpp]`
07. Contamination-Budgeted Frozen Rank Gate `[f360_determine_association_hypothesis.cpp]`
08. Doppler-Invariant Ghost Disambiguation `[f360_detect_multipath.cpp]`

Repository 2: Satellite Radar DSP Firmware

`Core_Radar_Gen8_iND13400`

09. Dual-Horizon Fast Dynamic Alignment `[dyn_align_wrapper.c]`
10. Range-Adaptive Dual-Loop Clutter CFAR `[appl_cfar_ifc.c]`
11. Occupancy-BLUE CFAR with Pfa Control `[appl_cfar_ifc.c / appl_adaptive_cfar.c]`
12. Priority-Ranked CDC Urban Packing `[cdc_packing.c / cdc_priority.c]`
13. Closed-Form Specular Contact Inversion `[ghost_calib.c]`
14. Track-Conditioned Residual Beat Estimator `[residual_track_ifc.c]`
15. 3-Product Cubic-Exact Residual Stencil `[residual_track_ifc.c]`

Repository 3: Domain Controller SiL Replay Harness

`Core_RESIM_DC_Emb_Library`

16. Preflight DA Header Injection `[sil_wrapper.cpp / Tracker_VariantA_Wrapper.cpp]`
17. Asynchronous Satellite Yaw Extrapolation `[sil_wrapper.cpp]`
18. Yaw-Free Timestamp and Doppler-Bias Separation `[sil_wrapper.cpp]`
19. Cross-Satellite Relative Yaw Cancellation `[sil_wrapper.cpp]`

Repository 4: Distributed Simulation and Cluster Infrastructure

`Core_RESIM_HPCC / Engine`

20. 4D Doppler-RadarSplat Generative Simulation `[models/doppler_radarsplat.py]`
21. Associative Parallel Prefix-Scan Kalman Filter `[engine/parallel_kf.py]`
22. Ghost-Conditioned Generative Target Augmentation `[models/ghost_augmentation.py]`
23. Nonlinear Parallel Gram-Lifted State Space `[engine/gram_lift.py]`
24. Guarded Error-Transfer Replay Certificates `[engine/guarded_replay.py]`

REPOSITORY 1

Central Object Tracker

`CORECOMP/ALSW/OT_ObjectTracking`

Research focus: Object tracking, multipath physics, association robustness, and uncertainty-aware filtering

8 concepts in this repository

Virtual Aperture Specular Tracking
Rank-1 Specular Invariant and Bearing Correction
Kinematics-Aware Adaptive Gating
Tangent-Mirror Hermite Linearization
Spectral Entropy Covariance Scaling
Distribution-Free Conformal Gating
Contamination-Budgeted Frozen Rank Gate
Doppler-Invariant Ghost Disambiguation

RESEARCH CONCEPT 01 / 24

Virtual Aperture Specular Tracking

Repository 1: Central Object Tracker | `Msmt_Update_ObjTrks_CCA_Moveable.cpp`

BASELINE / BEFORE

Single line-of-sight update

```
y_k = z_dir - h(x_k^-)
```

Multipath is penalized or suppressed through `mirror_prob`.

Tracking RMSE: 1.969 m.

PROPOSED / AFTER

Coupled-covariance bistatic update row

```
[y_dir; y_ghost] = [z_dir - h(x_k^-); z_ghost - h_s(x_k^-)]
R_eff = diag(sigma^2) + J_xi Sigma_xi J_xi^T
```

Tracking RMSE: 0.651 m, a 67.0% reduction in error.

DETAILED OPERATION AND MECHANISM

Pair validated specular reflections with their parent tracks. Append a synthetic observation row corresponding to a line of sight from the virtual sensor image s*. Propagate shared guardrail uncertainty Sigma_xi through the Jacobian J_xi and use a Joseph-form covariance update. This avoids treating direct and reflected observations as independent and prevents covariance collapse or filter overconfidence.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A host radar at (0, 0) tracks a car at (30, 2) m beside a guardrail at y = 3 m. The direct view has weak lateral resolution, approximately sigma_y = 1.8 m. Mirroring the sensor across the rail creates a virtual sensor at (0, 6) m, which observes the opposite flank of the car. The two viewing directions create cross-triangulation and reduce the lateral tracking error from 1.969 m to 0.651 m.

IMPLEMENTATION CHECKPOINTS

- Validate ghost-parent consistency before extending the measurement vector.
- Construct the virtual sensor pose from the estimated reflector plane.
- Augment innovation and measurement Jacobian with the specular row.
- Include reflector-state uncertainty in R_eff.
- Apply numerically stable Joseph covariance update and monitor positive definiteness.

KEY TECHNICAL RISK

Incorrect reflector association can inject a high-leverage synthetic row. Require temporal confirmation, innovation bounds, and covariance floors.

RESEARCH CONCEPT 02 / 24

Rank-1 Specular Invariant and Bearing Correction

Repository 1: Central Object Tracker | `f360_detect_multipath.cpp`

BASELINE / BEFORE

Target-mirror projection

```
p* = p - 2(n_hat^T p - d)n_hat
```

Range error: 0.0 m, but bearing error: 15.19 degrees, equivalent to an 8.0 m lateral bias in the example.

PROPOSED / AFTER

Virtual-sensor ray invariant

```
s* = 2d n_hat
p_s = p - 2d n_hat
||p - p_s - 2d n_hat|| <= 1.07e-14
```

Numerical invariant residual is effectively zero.

DETAILED OPERATION AND MECHANISM

Correct the arrival direction at the antenna array by expressing the path as a ray from the virtual sensor image to the real target. The baseline mirrors the target, which preserves total path length but reverses the arrival-bearing interpretation. The displacement invariant makes the geometry explicit and improves association between ghost detections and parent tracks.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

For a rail at y = 3 m with d = 3 and n_hat = [0, 1]^T, and a target at (30, 2) m, target mirroring gives (30, 4) m and bearing atan(4/30) = +7.59 degrees. The physical reflected wave is equivalent to a virtual sensor at (0, 6) m and therefore arrives along atan(-4/30) = -7.59 degrees. The baseline sign error is 15.19 degrees.

IMPLEMENTATION CHECKPOINTS

- Represent each planar reflector in Hessian normal form.
- Form the virtual sensor image, not a mirrored target, for bearing prediction.
- Use the invariant residual as a unit-test oracle.
- Preserve the correct coordinate-frame and angle-sign conventions.

KEY TECHNICAL RISK

Normal-vector sign and sensor-frame conventions can silently flip the result. Add coordinate-frame tests for all vehicle quadrants.

RESEARCH CONCEPT 03 / 24

Kinematics-Aware Adaptive Gating

Repository 1: Central Object Tracker | `f360_determine_association_hypothesis.cpp`

BASELINE / BEFORE

Fixed Mahalanobis threshold

```
d_M^2 = y_k^T S_k^-1 y_k <= gamma_0
gamma_0 = 9.21 (chi-square, 2 DoF, 99% bound)
```

Cut-in recall: 86.3%; maneuver RMSE: 0.283 m.

PROPOSED / AFTER

Dynamic kinematic expansion

```
gamma_ad = gamma_0 [1 + alpha ||a_target||/a_max + eta ||y_k||^2/Tr(S_k)]
Clamp gamma_ad to [9.21, 40.0].
```

Recall: 99.6%; maneuver RMSE: 0.212 m.

DETAILED OPERATION AND MECHANISM

Scale the association gate using target lateral acceleration and normalized innovation energy. A hard upper bound prevents uncontrolled spatial expansion in dense clutter, while the acceleration term preserves continuity during genuine cut-in or evasive maneuvers.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

For lateral acceleration 6.0 m/s^2, a_max = 8.0, alpha = 1.5, eta = 0.2, normalized innovation term 2.1, and d_M^2 = 18.4, the static 9.21 gate rejects the return. The adaptive threshold becomes approximately 23.4 and retains the detection.

IMPLEMENTATION CHECKPOINTS

- Estimate acceleration from a smoothed track state, not a single-frame difference.
- Normalize the innovation term to avoid unit dependence.
- Clamp both expansion and contraction.
- Log gate causes so KPI analysis can separate maneuver recovery from clutter admission.

KEY TECHNICAL RISK

Noisy acceleration estimates may expand gates spuriously. Apply persistence and acceleration-confidence checks.

RESEARCH CONCEPT 04 / 24

Tangent-Mirror Hermite Linearization

Repository 1: Central Object Tracker | `f360_is_reflective_guardrail_track.cpp`

BASELINE / BEFORE

Planar reflection or iterative grid search

```
Assume flat barrier: curvature kappa = 0
Alternative: nonlinear Fermat ray-tracing search
```

Curved-rail recall: 19.8%.

PROPOSED / AFTER

4 x 4 confluent Vandermonde solve

```
y(x) = a x^3 + b x^2 + c x + d
[y(x1), y'(x1), y(x2), y'(x2)] determines [a,b,c,d]
det(V) = (x2 - x1)^4 != 0
```

Coefficient reconstruction error: 3.44e-14.

DETAILED OPERATION AND MECHANISM

Intersect the direct-ghost perpendicular bisector with each receive ray to estimate two surface contact points. Convert local reflector normals into slopes, sigma = -n_x/n_y. Two points with two slopes provide Hermite constraints for a cubic barrier curve, solved non-iteratively.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A cloverleaf off-ramp rail curves from y = 3.0 m at x = 10 m to y = 5.2 m at x = 30 m. A planar model spreads ghost residuals along 132 m of rail. Two validated ghost pairs near x1 = 12 m and x2 = 24 m supply position and tangent constraints, reconstructing the local cubic to about 1e-14 numerical precision in under 0.1 ms.

IMPLEMENTATION CHECKPOINTS

- Reject nearly coincident x1 and x2 because the solve becomes ill-conditioned.
- Scale x coordinates before inversion.
- Use pivoted linear solve rather than explicit matrix inverse.
- Bound curvature and extrapolation distance outside the two contact points.

KEY TECHNICAL RISK

A cubic is local, not a globally valid road model. Restrict its use to the supported interval and uncertainty envelope.

RESEARCH CONCEPT 05 / 24

Spectral Entropy Covariance Scaling

Repository 1: Central Object Tracker | `Msmt_Update_ObjTrks_CTCA.cpp`

BASELINE / BEFORE

Static Gaussian measurement covariance

```
R_k = R_0 = diag(sigma_r^2, sigma_theta^2, sigma_v^2)
```

Wet-weather tracking RMSE: 0.911 m.

PROPOSED / AFTER

Entropy-modulated Joseph covariance

```
R(t) = R_0 exp(beta H(t)/SNR(t))
H(t) = - sum_i p_i ln(p_i)
```

Tracking RMSE: 0.315 m, a 65.4% reduction.

DETAILED OPERATION AND MECHANISM

Compute normalized Shannon entropy over Doppler-spectrum bins. Specular targets concentrate energy, whereas tire spray and diffuse surfaces distribute energy across many bins. As entropy rises relative to SNR, inflate measurement covariance. The Kalman gain falls and the tracker relies more strongly on its motion model.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

Behind a semi-truck in rain, returns span about +/-4 m/s with H = 2.85, SNR = 6 dB (linear 3.98), and beta = 4.0. The scale exp(4 x 2.85 / 3.98) is approximately 17.5. The reduced gain prevents heading oscillation driven by spray detections.

IMPLEMENTATION CHECKPOINTS

- Normalize spectral power into probabilities with a numerical floor.
- Convert SNR to a consistent linear or dB domain before applying the model.
- Cap covariance inflation and preserve minimum observability.
- Use Joseph form for covariance stability.

KEY TECHNICAL RISK

Low SNR alone can make the ratio unstable. Apply SNR floors and entropy confidence checks.

RESEARCH CONCEPT 06 / 24

Distribution-Free Conformal Gating

Repository 1: Central Object Tracker | `f360_determine_association_hypothesis.cpp`

BASELINE / BEFORE

Gaussian chi-square gating

```
Assume d_M^2 follows chi-square_d
```

Heavy-tailed maneuver coverage: 0.042; RMSE: 17.683 m.

PROPOSED / AFTER

Non-conformity quantile gate

```
G_(1-alpha) = {z : S(z) <= q_hat_(1-alpha)}
S_i = sqrt(y_i^T S_i^-1 y_i)
q_hat from empirical (1-alpha)(1+1/n) quantile
```

Empirical coverage: 99.6%; RMSE: 0.263 m.

DETAILED OPERATION AND MECHANISM

Maintain a calibration history of normalized innovation scores. Set the gate from the observed score quantile instead of a Gaussian theoretical quantile. This targets marginal empirical coverage even when residuals are skewed, multimodal, or heavy-tailed.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

During an S-curve slalom, a classical 9.21 threshold rejects 95 of 100 valid returns during transitions. The empirical residual distribution yields q_hat = 14.8 and retains 99.6% of valid detections.

IMPLEMENTATION CHECKPOINTS

- Use an exchangeable or approximately stationary calibration window.
- Separate operating regimes if residual distributions differ strongly.
- Define tie handling and finite-sample quantile indexing explicitly.
- Monitor realized coverage and gate width online.

KEY TECHNICAL RISK

Coverage guarantees weaken under severe temporal dependence or distribution shift. Combine with regime detection and bounded gates.

RESEARCH CONCEPT 07 / 24

Contamination-Budgeted Frozen Rank Gate

Repository 1: Central Object Tracker | `f360_determine_association_hypothesis.cpp`

BASELINE / BEFORE

Continuously sliding conformal threshold

`Online quantile updates continuously.`

Glint bursts inflate gate width from 1.58 to 30.0, admitting clutter.

PROPOSED / AFTER

Finite-budget rank sandwich

```
S_(231) <= q <= S_(247)
N = 256, replacement budget m = 8
Guaranteed lower coverage >= 0.9023
```

Gate remains bounded despite arbitrary bursts within the contamination budget.

DETAILED OPERATION AND MECHANISM

Freeze calibration on an independent reference block. Allow up to m adversarially replaced scores and bound the admissible threshold between protected order statistics. Add a 2-of-3 sensor quorum before accepting gate adaptation.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A bridge expansion joint creates six glint spikes with d_M^2 above 100. A sliding gate widens to 30.0. The frozen rank gate reserves the upper eight positions and anchors the threshold near rank 247, keeping the effective boundary at or below 2.14 in the stated score scale.

IMPLEMENTATION CHECKPOINTS

- Select a clean reference interval and version it with the calibration.
- Choose the replacement budget from measured burst statistics.
- Do not update the frozen window during an active contamination alarm.
- Use quorum logic to avoid a single-sensor poisoning event.

KEY TECHNICAL RISK

A stale reference block may under-cover a real regime change. Include controlled re-calibration criteria.

RESEARCH CONCEPT 08 / 24

Doppler-Invariant Ghost Disambiguation

Repository 1: Central Object Tracker | `f360_detect_multipath.cpp`

BASELINE / BEFORE

Position-only geometric association

`Associate candidate reflection using distance and geometry only.`

Decoy precision: 0.638; false positives: 36.2%.

PROPOSED / AFTER

Radial Doppler projection gate

`|v_r,meas - u_spec^T v| <= 3 sqrt(sigma_v^2 + S_v)`

`Require confirmation in two scans.`

Decoy precision: 0.877; ghost-rejection rate: 66.1%.

DETAILED OPERATION AND MECHANISM

Project the tracked target velocity onto the predicted specular line of sight. Compare the predicted radial speed with each candidate return. Stationary roadside objects may match reflected position geometry but generally fail the Doppler consistency test.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A vehicle moves at 25 m/s along a wall. The direct radar measures -24.8 m/s. A road sign at a similar mirrored distance measures -18.2 m/s, while the specular model predicts -20.19 m/s. The 1.987 m/s mismatch is reported as roughly 40 sigma under the stated noise model, so the sign is rejected.

IMPLEMENTATION CHECKPOINTS

- Propagate track velocity covariance into S_v.
- Use consistent radial-velocity signs and ego-motion compensation.
- Require two-scan confirmation to suppress isolated Doppler outliers.
- Relax the test at near-zero relative Doppler where discrimination is weak.

KEY TECHNICAL RISK

Velocity bias or poor ego-motion compensation can reject real ghosts. Calibrate Doppler bias before enabling a tight gate.

REPOSITORY 2

Satellite Radar DSP Firmware

`Core_Radar_Gen8_iND13400`

Research focus: Embedded radar calibration, CFAR, prioritization, and low-compute estimators

7 concepts in this repository

Dual-Horizon Fast Dynamic Alignment
Range-Adaptive Dual-Loop Clutter CFAR
Occupancy-BLUE CFAR with Pfa Control
Priority-Ranked CDC Urban Packing
Closed-Form Specular Contact Inversion
Track-Conditioned Residual Beat Estimator
3-Product Cubic-Exact Residual Stencil

RESEARCH CONCEPT 09 / 24

Dual-Horizon Fast Dynamic Alignment

Repository 2: Satellite Radar DSP Firmware | `dyn_align_wrapper.c`

BASELINE / BEFORE

Static gain schedule

`K(N) = K_0 approximately 0.0025`

`Startup deadband: N_wait = 400 frames`

Lock may require 20 s at 20 Hz, and substantially longer when road-curve pauses are applied.

PROPOSED / AFTER

Dual-horizon exponential decay

`K_DA(N) = K_fast exp(-alpha N) + K_slow`

`K_fast = 0.05, alpha = 0.08, K_slow = 0.0025`

Convergence in at most 46 frames, below 2.3 s at 20 Hz.

DETAILED OPERATION AND MECHANISM

Use a high initial correction gain and exponentially decay toward the stable production gain. Apply a stationary-clutter Doppler RANSAC gate, |v_r + v_ego cos(theta)| <= epsilon, so turns and moving objects do not corrupt boresight estimation.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A vehicle exits a garage with azimuth error -1.145 degrees. The baseline requires 400 accepted frames; road-curvature pauses can extend elapsed time. The dual-horizon schedule makes large changes in frames 1-30 and locks when residual variance stabilizes at frame 46.

IMPLEMENTATION CHECKPOINTS

- Cap startup updates in Auto_align_Init().
- Gate candidates with ego-compensated stationary Doppler.
- Switch or decay gain based on both frame count and residual stability.
- Persist converged alignment with validity metadata.

KEY TECHNICAL RISK

Aggressive startup gain can chase moving clutter. Require spatial diversity and robust inlier selection.

RESEARCH CONCEPT 10 / 24

Range-Adaptive Dual-Loop Clutter CFAR

Repository 2: Satellite Radar DSP Firmware | `appl_cfar_ifc.c`

BASELINE / BEFORE

Static-reference-window CA-CFAR

```
T(r) = alpha_CA / R_0 * sum_{i in W} P(r+i)
R_0 = 16
```

Edge recall: 50.0%; false alarms: 1 in the stated test.

PROPOSED / AFTER

Dual-loop background smoothing

```
C_hat_(k+1)(r) = (1-lambda) C_hat_k(r) +
lambda median_d P(r,d)
s(r) = 0.5 |C_hat(r+1) - C_hat(r-1)|
R_0 in {8,16,32}, with clean-side censoring
```

Recall: 100.0%; false alarms: 0 in the stated test.

DETAILED OPERATION AND MECHANISM

Maintain a slow background clutter floor using track-free Doppler bins. Detect abrupt range gradients at guardrail or wall edges, shrink the reference window, and censor cells on the contaminated side so the threshold is estimated from clean background.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

At 1.2 m from a metal guardrail, a +45 dB return spills across six range bins. Static CA-CFAR raises the threshold by 18 dB and misses the nearby car. The adaptive method detects the slope, chooses R_0 = 8, excludes wall-facing cells, and recovers the target.

IMPLEMENTATION CHECKPOINTS

- Update the background using only bins without active tracks or strong detections.
- Use hysteresis when changing window size.
- Maintain sufficient clean cells for stable Pfa control.
- Expose edge-state diagnostics for validation.

KEY TECHNICAL RISK

Over-censoring reduces the statistical sample size and may destabilize Pfa. Enforce minimum reference-cell counts.

RESEARCH CONCEPT 11 / 24

Occupancy-BLUE CFAR with Pfa Control

Repository 2: Satellite Radar DSP Firmware | `appl_cfar_ifc.c / appl_adaptive_cfar.c`

BASELINE / BEFORE

Uniform cell weighting

```
beta_i = 1/N
```

Reference cells are treated equally or removed by binary censoring, reducing sensitivity when partially occupied.

PROPOSED / AFTER

Variance-optimal BLUE weighting

```
beta_i proportional to m_i / v_i
Pfa(alpha) = product_i (1 + alpha beta_i)^-1
```

Target Pfa 1e-4 gives 1.02e-4; weak-target recall improves by 7.4%.

DETAILED OPERATION AND MECHANISM

Use occupancy probability and expected cell statistics to soft-weight each reference cell. Solve the closed-form product equation for alpha by bounded bisection, retaining constant false-alarm behavior without large heuristic lookup tables.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A 16-cell reference window overlaps two adjacent-vehicle cells with occupancy probability 0.85. Uniform averaging lets leaked target power inflate the noise estimate. BLUE assigns those cells a much smaller effective weight, stated as 0.08 relative to 1.0 for clean cells, improving weak pedestrian recall by 7.4%.

IMPLEMENTATION CHECKPOINTS

- Normalize weights and bound extreme ratios.
- Solve alpha monotonically with fixed iteration limits suitable for firmware.
- Validate fixed-point quantization error in Pfa.
- Fallback to conventional CFAR if occupancy metadata is unavailable.

KEY TECHNICAL RISK

Miscalibrated occupancy probabilities can bias the threshold. Track Pfa by scene class and recalibrate weights.

RESEARCH CONCEPT 12 / 24

Priority-Ranked CDC Urban Packing

Repository 2: Satellite Radar DSP Firmware | `cdc_packing.c / cdc_priority.c`

BASELINE / BEFORE

First-come, first-served truncation

`Sequentially fill CDC output until MAX_CDC = 5016.`

Under saturation, late detections are dropped regardless of safety relevance.

PROPOSED / AFTER

Threat-weighted top-k selection

`score = SNR_linear / r^2`

`In-place O(N*k) selection for bounded firmware memory.`

Near-field and high-risk detections are packed before low-priority background.

DETAILED OPERATION AND MECHANISM

Score range-Doppler candidates before Ethernet packaging. Reorder p_cdc_array and p_ci_flag_array with the same permutation, then serialize the highest-ranked detections into each 1472-byte UDP payload first.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A crowded intersection yields 6,200 detections, exceeding the 5,016 limit. Baseline truncation misses a pedestrian at 4 m. Its score is 31.62/16 = 1.98 for 15 dB SNR, while an 85 m building at 30 dB scores 1000/7225 = 0.138. The pedestrian is retained.

IMPLEMENTATION CHECKPOINTS

- Use stable tie-breaking to keep replay deterministic.
- Apply floors to range and caps to SNR.
- Keep metadata arrays synchronized during swaps.
- Measure worst-case CPU time at maximum N and k.

KEY TECHNICAL RISK

The simple score may over-prioritize benign near-field clutter. Add class, closing speed, and confidence when available.

RESEARCH CONCEPT 13 / 24

Closed-Form Specular Contact Inversion

Repository 2: Satellite Radar DSP Firmware | `ghost_calib.c`

BASELINE / BEFORE

Iterative ray tracing

Numerically solve a nonlinear Fermat-path objective.

Iterative searches can exceed a 50 ms real-time cycle.

PROPOSED / AFTER

Closed-form algebraic solution

```
t = 2(l - u^T d) / (l^2 - ||d||^2)
Curvature certificate: ||g''|| <= 2 kappa_max rho_max
```

Contact-point RMSE: 0.018 m.

DETAILED OPERATION AND MECHANISM

Solve the physical bounce point directly along receive unit vector u using path-length geometry. Evaluate a curvature limit that certifies whether the local planar-reflector assumption is valid; reject rather than force a solution outside that bound.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

For target p = (20, 2) m, sensor at the origin, wall distance 3.5 m, and total path length 28.42 m, the iterative baseline needs about 15 Newton steps. The closed form gives t = 8.12 m with six stated floating-point operations and 0.018 m error.

IMPLEMENTATION CHECKPOINTS

- Check denominator magnitude and physical range of t.
- Verify path-length closure after solving.
- Evaluate the curvature certificate before accepting calibration.
- Use double precision offline to generate fixed-point error bounds.

KEY TECHNICAL RISK

Degenerate geometries can make the denominator small. Detect ill-conditioning and fall back safely.

RESEARCH CONCEPT 14 / 24

Track-Conditioned Residual Beat Estimator

Repository 2: Satellite Radar DSP Firmware | `residual_track_ifc.c`

BASELINE / BEFORE

Full 512-point range FFT

```
X[k] = sum_{n=0}^{N-1} x[n] W_N^(nk)
```

4,608 DSP multiplications per dwell; floating-point range RMSE: 0.439 m.

PROPOSED / AFTER

Lag-1 autocorrelation residual

```
f_res = angle(sum_{n=1}^{N-1} z[n]
conjugate(z[n-1])))/(2 pi T_s)
```

1,024 multiplications, 4.5x compute reduction; INT8 RMSE: 0.108 m.

DETAILED OPERATION AND MECHANISM

Derotate raw fast-time ADC samples using the track-predicted center frequency. Estimate only the small residual frequency from the average lag-1 phase, enabling sub-bin refinement with low arithmetic cost and bounded INT8 implementation.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

For a vehicle at 42.3 m with predicted beat frequency 1.41 MHz, the baseline runs a 512-point FFT plus interpolation. The residual method derotates the samples and evaluates lag-1 correlation in 1,024 operations, reaching 0.108 m RMSE under INT8 arithmetic in the stated test.

IMPLEMENTATION CHECKPOINTS

- Gate use by track confidence and predicted-bin error.
- Accumulate products with sufficient headroom.
- Handle phase wrapping and low-coherence cases.
- Retain a full FFT fallback for track acquisition.

KEY TECHNICAL RISK

A bad predicted frequency can move the residual outside its unambiguous interval. Detect and revert to coarse FFT.

RESEARCH CONCEPT 15 / 24

3-Product Cubic-Exact Residual Stencil

Repository 2: Satellite Radar DSP Firmware | `residual_track_ifc.c`

BASELINE / BEFORE

Polynomial phase fitting

`Fit phase coefficients by least squares across many ADC samples.`

Accurate but requires matrix operations and additional samples.

PROPOSED / AFTER

Three-complex-product stencil

```
eta_0 = angle(z_2 conjugate(z_1))
eta_+ = angle(z_3 conjugate(z_2))
eta_- = angle(z_1 conjugate(z_0))
f_hat = (6 eta_0 - eta_+ - eta_-)/(24 h)
```

Range RMSE: 13.13 mm with three complex products.

DETAILED OPERATION AND MECHANISM

Use four lag-separated samples and three adjacent complex products. A fixed linear combination cancels modeled quadratic and cubic phase curvature, leaving the carrier-frequency term while avoiding least-squares inversion.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A vehicle accelerating at 8 m/s^2 adds chirp curvature and spreads a conventional estimator over about two frequency bins. The stencil cancels the modeled higher-order components and estimates the center frequency within 13.13 mm of range truth in the stated test.

IMPLEMENTATION CHECKPOINTS

- Derive coefficients against the exact phase convention used in firmware.
- Select lag h to balance noise and phase wrapping.
- Use circular phase handling.
- Validate fixed-point coefficient quantization.

KEY TECHNICAL RISK

The exactness depends on the assumed polynomial order and sample timing. Jerk or timing jitter can introduce bias.

REPOSITORY 3

Domain Controller SiL Replay Harness

`Core_RESIM_DC_Emb_Library`

Research focus: Deterministic replay, asynchronous sensor deskew, and calibration invariants

4 concepts in this repository

Preflight DA Header Injection
Asynchronous Satellite Yaw Extrapolation
Yaw-Free Timestamp and Doppler-Bias Separation
Cross-Satellite Relative Yaw Cancellation

RESEARCH CONCEPT 16 / 24

Preflight DA Header Injection

Repository 3: Domain Controller SiL Replay Harness | `sil_wrapper.cpp / Tracker_VariantA_Wrapper.cpp`

BASELINE / BEFORE

Uninitialized cold start

`theta_init = 0.0 degrees`

Slice boundaries can trigger repeated convergence delays; the supplied estimate states 360,000 frames lost across 20 HPCC nodes.

PROPOSED / AFTER

Header-stream angle seeding

`theta_0 = theta_header at t = 0`

`Read _DYNAMIC_ALIGNMENT_STREAM into M2D_One_Time_Msg_T.`

Startup latency drops to zero frames when a valid header value exists.

DETAILED OPERATION AND MECHANISM

Read converged azimuth and elevation alignment from the log header and inject it into the radar configuration structure before frame 1. Each distributed replay slice begins from the recorded calibration state rather than replaying the convergence transient.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

With a 300-log drive split over 20 nodes, Node 2 starts at Log 16. Baseline initialization uses 0.0 degrees and needs 400 frames, failing early UDP angle KPIs. Header injection seeds -1.145 degrees immediately.

IMPLEMENTATION CHECKPOINTS

- Validate stream presence, version, endianness, and validity flags.
- Inject before any measurement-processing callback.
- Record the seeded value in replay metadata.
- Fallback explicitly to cold-start behavior when the header is absent.

KEY TECHNICAL RISK

A stale or incompatible header can hide real alignment behavior. Include source timestamp and calibration version checks.

RESEARCH CONCEPT 17 / 24

Asynchronous Satellite Yaw Extrapolation

Repository 3: Domain Controller SiL Replay Harness | `sil_wrapper.cpp`

BASELINE / BEFORE

Uncompensated timestamp stacking

```
p_fused = p_meas
Delta t in [5,45] ms
```

Coordinate error: 0.627 m to 0.711 m.

PROPOSED / AFTER

Kinematic rigid-body deskewing

```
p_corr = R_z(omega Delta t) p_meas - v_ego
Delta t - 0.5 a_ego Delta t^2
```

Fusion error reduction: 86.2% at omega = 0.1 rad/s and 31.7% at omega = 0.6 rad/s.

DETAILED OPERATION AND MECHANISM

Transform each satellite detection from its measurement epoch to the central Domain Controller epoch. Apply yaw rotation plus translational ego-motion compensation to eliminate duplicated or smeared targets during turns.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

At 45 km/h in a roundabout with yaw rate 0.4 rad/s and 35 ms sensor skew, front-left and front-right detections of one vehicle differ by 0.68 m. Deskewing aligns them to about 0.09 m error.

IMPLEMENTATION CHECKPOINTS

- Define a single reference timestamp and frame convention.
- Interpolate ego yaw rate, velocity, and acceleration to each measurement time.
- Propagate timing and ego-motion uncertainty into detection covariance.
- Unit-test positive and negative Delta t.

KEY TECHNICAL RISK

High yaw-rate motion violates a constant-rate approximation over long delays. Use higher-order integration or shorter buffering when necessary.

Yaw-Free Timestamp and Doppler-Bias Separation

Repository 3: Domain Controller SiL Replay Harness | `sil_wrapper.cpp`

BASELINE / BEFORE

Coupled nonlinear estimation

`Jointly estimate time offset tau, Doppler bias b, and mounting yaw beta.`

The parameters can become confounded during acceleration and weak excitation.

PROPOSED / AFTER

Independent linear Gram invariant

`r_k w_k - d_k^T v_k = ||v_k||^2 tau`

Separates tau and b to about 1e-12 numerical precision; timing RMSE: 3.8 ms.

DETAILED OPERATION AND MECHANISM

Combine measured range and range rate with reference trajectory vectors so the unknown mounting-yaw rotation cancels from the timing equation. Solve a two-target Gram system for clock offset and Doppler bias, then estimate yaw after temporal correction.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A satellite radar has 15 ms clock delay and 0.8 degree yaw error. The nonlinear baseline trades one error against the other. The invariant equation removes yaw from the timing solve and recovers the 15 ms offset independently.

IMPLEMENTATION CHECKPOINTS

- Require targets with sufficiently independent velocity vectors.
- Check the Gram matrix condition number.
- Estimate Doppler bias jointly only after units and signs are normalized.
- Solve yaw after timestamp correction.

KEY TECHNICAL RISK

Poor target geometry makes the Gram system singular or noisy. Add excitation checks before calibration.

Cross-Satellite Relative Yaw Cancellation

Repository 3: Domain Controller SiL Replay Harness | `sil_wrapper.cpp`

BASELINE / BEFORE

Clutter-dependent alignment

`Estimate boresight from stationary roadside features.`

Feature-sparse highways can leave drift unobserved by up to 13.0 degrees.

PROPOSED / AFTER

Direct-ghost difference phase

```
D_i,k = exp(-j beta_i) (p_k - p*_k)
delta_hat_ij = arg(sum_k D_i,k conjugate(D_j,k))
```

Relative boresight yaw recovered to 1e-10 rad in the stated synthetic test.

DETAILED OPERATION AND MECHANISM

For overlapping sensors, subtract direct and ghost vectors. This cancels common target motion and much of the wall geometry. Cross-correlating the complex difference vectors isolates the relative sensor rotation.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

On an open desert highway without poles or signs, independent alignment drifts. Front-right and rear-right sensors observe a passing vehicle and its rail reflection. Cross-multiplication cancels the trajectory and estimates relative yaw without stationary point clutter.

IMPLEMENTATION CHECKPOINTS

- Confirm that direct and ghost observations share the same parent.
- Accumulate across multiple frames and weight by covariance.
- Reject low-magnitude difference vectors.
- Anchor the relative network with one absolute calibration source.

KEY TECHNICAL RISK

Incorrect ghost pairing directly corrupts relative yaw. Use strict temporal and Doppler consistency.

REPOSITORY 4

Distributed Simulation and Cluster Infrastructure

`Core_RESIM_HPCC / Engine`

Research focus: Radar-native simulation, parallel state estimation, synthetic augmentation, and replay certification

5 concepts in this repository

4D Doppler-RadarSplat Generative Simulation
Associative Parallel Prefix-Scan Kalman Filter
Ghost-Conditioned Generative Target Augmentation
Nonlinear Parallel Gram-Lifted State Space
Guarded Error-Transfer Replay Certificates

RESEARCH CONCEPT 20 / 24

4D Doppler-RadarSplat Generative Simulation

Repository 4: Distributed Simulation and Cluster Infrastructure | `models/doppler_radarsplat.py`

BASELINE / BEFORE

Optical image synthesis

Render RGB or depth approximations; omit Doppler, RCS, and CDC spectra.

Insufficient for firmware-level Range-Doppler validation.

PROPOSED / AFTER

Differentiable RF primitives

```
v_r,i = (v_i - v_ego)^T u_i
P_r proportional to sigma_RCS / R^4
```

Analytic gradient agreement below 1e-8; velocity RMSE: 0.036 m/s in the stated comparison.

DETAILED OPERATION AND MECHANISM

Extend dynamic Gaussian splats with radar cross section, range-dependent attenuation, line-of-sight Doppler, and spectral spread. Rasterize them into synthetic Range-Doppler compressed data cubes suitable for firmware and tracker testing.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A car travels at 30 m/s at 40 m range. Optical simulators output shape and depth only. RadarSplat applies R^-4 power loss and radial projection to synthesize a CDC spectrum whose velocity matches logs within 0.036 m/s RMSE in the stated test.

IMPLEMENTATION CHECKPOINTS

- Keep all RF operations differentiable for parameter fitting.
- Model antenna pattern and occlusion, not only R^-4 loss.
- Calibrate RCS distributions by class and aspect.
- Validate both aggregate KPIs and individual spectral signatures.

KEY TECHNICAL RISK

A visually plausible synthetic scene can still be electromagnetically wrong. Validate against measured CDC distributions.

Associative Parallel Prefix-Scan Kalman Filter

Repository 4: Distributed Simulation and Cluster Infrastructure | **engine/parallel_kf.py**

BASELINE / BEFORE

Serial time-stepping loop

```
x_k = F_k x_(k-1) + K_k y_k
```

50,000 frames require 50,000 dependent steps and form a replay bottleneck.

PROPOSED / AFTER

Associative binary-tree scan

```
S_(i,j) = S_(i,k) tensor S_(k+1,j)
50,000 steps map to 16 parallel levels.
```

Span reduction: 3,125x; relative numerical error: 6.35e-15.

DETAILED OPERATION AND MECHANISM

Represent each filtering interval as an associative information-form element (A, b, C, eta, J). Combine adjacent intervals using a Blelloch prefix scan so GPU workers process a binary tree instead of a serial chain.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A 50,000-frame vehicle track takes 45 s in a serial replay. The stated GPU prefix version processes 16 tree levels in under 0.02 s while matching serial EKF position to 1e-10 m.

IMPLEMENTATION CHECKPOINTS

- Prove and unit-test associativity for the exact element representation.
- Use stable matrix factorizations and symmetry enforcement.
- Support segmented scans for track births and deaths.
- Compare against a serial oracle across random seeds.

KEY TECHNICAL RISK

Numerical associativity is approximate in floating point. Reduction order and ill-conditioned covariances can affect reproducibility.

RESEARCH CONCEPT 22 / 24

Ghost-Conditioned Generative Target Augmentation

Repository 4: Distributed Simulation and Cluster Infrastructure | `models/ghost_augmentation.py`

BASELINE / BEFORE

Uniform synthetic insertion

`Insert reflections without dynamic-range balancing; stated power spread is about 500x.`

False-alarm rate: 0.425.

PROPOSED / AFTER

Physics-normalized ghost injection

`Generate specular geometry and normalize power with tracker-predicted range and reflector properties.`

False-alarm rate: 0.033; recall: 1.000.

DETAILED OPERATION AND MECHANISM

Generate multipath geometry inside the Doppler-RadarSplat pipeline. Scale reflection energy using range, incidence angle, guardrail conductivity, and predicted target power so the synthetic distribution does not overwhelm the learner.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

Targets at 5 m can be about 500 times stronger than targets at 70 m under the chosen scenario. Uniform insertions teach a detector to fire on synthetic artifacts, producing 42.5% false alarms. Physics normalization reduces false alarms to 3.3% while preserving 100% recall.

IMPLEMENTATION CHECKPOINTS

- Condition on scene geometry, reflector material, and target state.
- Match real-log amplitude and entropy distributions.
- Separate training augmentation from validation data generation.
- Track performance by range and ghost type.

KEY TECHNICAL RISK

Normalization can accidentally leak tracker labels into training. Keep causal feature boundaries explicit.

RESEARCH CONCEPT 23 / 24

Nonlinear Parallel Gram-Lifted State Space

Repository 4: Distributed Simulation and Cluster Infrastructure | **engine/gram_lift.py**

BASELINE / BEFORE

Iterated extended Kalman smoothing

Repeatedly relinearize nonlinear measurements around a trajectory.

Outer iterations create sequential dependencies across sweeps.

PROPOSED / AFTER

Quadratic invariant state lifting

`zeta = [||p||^2, p^T v, ||v||^2]^T`

Use Pascal-matrix transitions for polynomial kinematics.

Linear measurement closure error: 5.87e-13.

DETAILED OPERATION AND MECHANISM

Augment the state with quadratic inner-product invariants. Range-squared and range-rate relations become linear in the lifted state, allowing direct use within an associative prefix scan without outer relinearization loops.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A maneuvering vehicle is observed through range, bearing, and radial speed. Conventional parallel smoothing still requires trajectory-dependent relinearization. The lifted invariants make range-squared updates linear and preserve parallel execution.

IMPLEMENTATION CHECKPOINTS

- Define closure for every transition and process-noise term.
- Control lifted-state covariance growth.
- Project back to physically consistent p and v states.
- Compare lifted and iterated solutions on strongly nonlinear cases.

KEY TECHNICAL RISK

State lifting increases dimension and may admit inconsistent invariant combinations. Use constraint projection or regularization.

RESEARCH CONCEPT 24 / 24

Guarded Error-Transfer Replay Certificates

Repository 4: Distributed Simulation and Cluster Infrastructure | `engine/guarded_replay.py`

BASELINE / BEFORE

Empirical warm-up windows

Prepend a guessed warm-up interval, for example 200 frames.

State and track-ID divergence remains possible near slice boundaries.

PROPOSED / AFTER

Guarded scalar affine operators

Represent interval as S = (a, b, h).

A certified radius e < 1/3 preserves discrete branch equivalence.

Enables slice equivalence checks without executing arbitrary warm-up frames.

DETAILED OPERATION AND MECHANISM

Summarize an interval using affine error transfer and explicit distances to branch boundaries, such as track-birth or deletion thresholds. If propagated error remains inside the guard margin, the sliced replay is certified to make the same discrete decisions as continuous execution.

CONCRETE NUMERICAL / PHYSICAL EXAMPLE

A log is split at frame 10,000. A track born at 9,950 may not be reconstructed by a guessed 200-frame warm-up if prior state differs. The certificate evaluates the decision guard, stated as $h_k > 0.33$, and proves whether the slice preserves the continuous birth/death sequence.

IMPLEMENTATION CHECKPOINTS

- Enumerate every discrete branch that affects externally visible state.
- Compute conservative error-transfer bounds.
- Fail closed when a guard cannot be certified.
- Store certificate results alongside replay KPIs.

KEY TECHNICAL RISK

A certificate is only as complete as the modeled branches. Unmodeled saturation, quantization, or nondeterminism can invalidate it.

Integrated Architecture and Recommended Validation

The concepts form a coherent stack: radar-native data generation and firmware estimation feed time-aligned domain-controller replay; robust association and multipath-aware tracking consume those measurements; the HPCC engine accelerates large-scale evaluation and certifies slice equivalence.

END-TO-END INFORMATION FLOW

- 1. Scene and RF generation: Doppler-RadarSplat plus ghost-conditioned augmentation.
- 2. Firmware extraction: adaptive CFAR, residual beat estimation, calibration, and priority packing.
- 3. Replay conditioning: alignment seeding, timestamp deskew, bias separation, relative yaw.
- 4. Object tracking: specular geometry, adaptive/conformal gating, entropy-aware covariance.
- 5. Cluster validation: prefix scans, lifted state models, and guarded replay certificates.

Minimum validation package for each proposal

- A written hypothesis and failure mode tied to one target file.
- A serial or baseline oracle with frozen inputs and deterministic outputs.
- Unit tests for geometry, signs, frames, limits, and degenerate cases.
- Scenario matrix covering nominal, maneuver, clutter, multipath, weather, and saturation.
- KPIs with numerator/denominator definitions, confidence intervals, and scene stratification.
- Runtime, memory, fixed-point, and worst-case execution-time evidence where firmware is affected.
- Ablation study showing that the claimed gain comes from the proposed mechanism.
- Versioned artifacts: commit, configuration, log list, seed, command, and KPI output.

Recommended prioritization

Wave	Candidate concepts	Why now	Exit criterion
1	Header injection; yaw deskew; bearing correction; CDC ranking	High value and relatively bounded integration	Deterministic replay and unit-tested frame/sign behavior
2	Adaptive gating; entropy covariance; Doppler ghost gate; residual estimator	Direct KPI gains with manageable algorithmic risk	Scene-stratified KPI improvement without FA regression
3	Conformal/frozen gates; advanced CFAR; fast alignment	Requires calibration governance and robust-statistics validation	Coverage/Pfa targets sustained under contamination and shift
4	RadarSplat; prefix KF; Gram lift; replay certificates	Transformational but cross-repository and infrastructure-heavy	End-to-end equivalence, scaling, and measured-log fidelity

Appendix A: Reported Metrics Index

This compact index supports review planning. Values are reported from the supplied notes and should be traced to reproducible evidence.

#	Concept	Repo	Reported result
1	**Virtual Aperture Specular Tracking**	1	Tracking RMSE: 0.651 m, a 67.0% reduction in error.
2	**Rank-1 Specular Invariant and Bearing Correction**	1	Numerical invariant residual is effectively zero.
3	**Kinematics-Aware Adaptive Gating**	1	Recall: 99.6%; maneuver RMSE: 0.212 m.
4	**Tangent-Mirror Hermite Linearization**	1	Coefficient reconstruction error: 3.44e-14.
5	**Spectral Entropy Covariance Scaling**	1	Tracking RMSE: 0.315 m, a 65.4% reduction.
6	**Distribution-Free Conformal Gating**	1	Empirical coverage: 99.6%; RMSE: 0.263 m.
7	**Contamination-Budgeted Frozen Rank Gate**	1	Gate remains bounded despite arbitrary bursts within the contamination budget.
8	**Doppler-Invariant Ghost Disambiguation**	1	Decoy precision: 0.877; ghost-rejection rate: 66.1%.
9	**Dual-Horizon Fast Dynamic Alignment**	2	Convergence in at most 46 frames, below 2.3 s at 20 Hz.
10	**Range-Adaptive Dual-Loop Clutter CFAR**	2	Recall: 100.0%; false alarms: 0 in the stated test.
11	**Occupancy-BLUE CFAR with Pfa Control**	2	Target Pfa 1e-4 gives 1.02e-4; weak-target recall improves by 7.4%.
12	**Priority-Ranked CDC Urban Packing**	2	Near-field and high-risk detections are packed before low-priority background.
13	**Closed-Form Specular Contact Inversion**	2	Contact-point RMSE: 0.018 m.
14	**Track-Conditioned Residual Beat Estimator**	2	1,024 multiplications, 4.5x compute reduction; INT8 RMSE: 0.108 m.
15	**3-Product Cubic-Exact Residual Stencil**	2	Range RMSE: 13.13 mm with three complex products.
16	**Preflight DA Header Injection**	3	Startup latency drops to zero frames when a valid header value exists.
17	**Asynchronous Satellite Yaw Extrapolation**	3	Fusion error reduction: 86.2% at omega = 0.1 rad/s and 31.7% at omega = 0.6 rad/s.
18	**Yaw-Free Timestamp and Doppler-Bias Separation**	3	Separates tau and b to about 1e-12 numerical precision; timing RMSE: 3.8 ms.
19	**Cross-Satellite Relative Yaw Cancellation**	3	Relative boresight yaw recovered to 1e-10 rad in the stated synthetic test.
20	**4D Doppler-RadarSplat Generative Simulation**	4	Analytic gradient agreement below 1e-8; velocity RMSE: 0.036 m/s in the stated comparison.
21	**Associative Parallel Prefix-Scan Kalman Filter**	4	Span reduction: 3,125x; relative numerical error: 6.35e-15.
22	**Ghost-Conditioned Generative Target Augmentation**	4	False-alarm rate: 0.033; recall: 1.000.
23	**Nonlinear Parallel Gram-Lifted State**	4	Linear measurement closure error: 5.87e-13.

#	Concept	Repo	Reported result
	Space		
24	**Guarded Error-Transfer Replay Certificates**	4	Enables slice equivalence checks without executing arbitrary warm-up frames.